Module-Wrestler Online Character Dictionary

Gabe Field

October 21, 2026

Abstract

We are the most modern iteration of a long-established translation dictionary relating the language of characters with the language of finite complex group modules.

1 The Translation Dictionary

There are categories **FinGroup** of finite groups, **FinVect** of finite-dimensional vector spaces over $\mathbb{C}$, and for each $G : \mathbf{FinGroup}$, a category $\mathbf{FinVect}^G$ of finite-dimensional complex G-modules. If $V : \mathbf{FinVect}^G$, the G-action will be denoted $G \xrightarrow{\rho_V} \mathrm{GL}(V)$ or $g \cdot x := \rho_V(g)(x)$. If it is clear, we may denote $\rho_V(g)$ by $g \cdot -$. The set of conjugacy classes will be denoted $\mathcal{CC}(G)$. For $x \in G$, its conjugacy class is denoted $\mathcal{CL}(x)$.

1.1 Definitions in the Language of Characters

Throughout, $G : \mathbf{FinGroup}$ is a group and $V : \mathbf{FinVect}^G$ is a finite-dimensional complex G-module. (Any other type data will be specified.)

(D1) **Character** of V: The function $\chi_V := \mathrm{tr} \circ \rho_V : G \to \mathbb{C}$. It is a class function (D3).

(D2) **Character table** of G: A listing of the characters χ_I corresponding to the finite-dimensional complex irreducible modules I of G. As characters are class functions (D3), the value on each *conjugacy class* is listed. As the irreducible modules biject with the conjugacy classes, this is a square matrix. For example, behold the character table of S_2 (it's not interesting):

Character	$(*)(*)$	$(**)$
χ_{trivial}	1	1
χ_{sign}	1	-1

Beware – although the row for χ_I encodes the character χ_I, it is missing information on the size of the conjugacy class. This information is critical to computing the inner product of two characters (D4); generically, the inner product **does not** agree with the "dot product".

(D3) **Class function**: A function $\phi : G \to A$ which is constant on conjugacy classes of G. A is typically $\mathbb{C}$.

(D4) **Inner product** on $\mathbb{C}^G$: The vector space of functions $G \to \mathbb{C}$ carries Hermitian inner product $\langle -, - \rangle$ given by

$$\langle \phi, \psi \rangle := \frac{1}{|G|} \sum_{g \in G} \phi(g) \cdot \overline{\psi(g)}$$

or, for characters $(U, V : \mathbf{FinVect}^G)$:

$$\langle \chi_U, \chi_V \rangle := \frac{1}{|G|} \sum_{g \in G} \chi_U(g) \cdot \chi_V(g^{-1})$$

because of (G1).

(D5) **Dimension/Degree** of V: The dimension $\dim(V)$.

1.2 Grammatical Rules of the Language of Characters

I mean in this section to list some key properties that characters always satisfy. If you were to communicate using characters, these are some rules that "always appear" in your sentences! Throughout, $G : \mathbf{FinGroup}$ and $U, V : \mathbf{FinVect}^G$ unless otherwise specified.

(G1) **Inverses**: For $g \in G$, $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.

(G2) **Frobenius reciprocity**: If $H \leq G$, $U : \mathbf{FinVect}^H$ and $V : \mathbf{FinVect}^G$, then

$$\langle \mathrm{Ind}_H^G U, V \rangle = \langle U, \mathrm{Res}_H^G V \rangle.$$

See (T1).

(G3) **Orthonormality of irreducibles**:

$$U, V \text{ both irreducible} \implies \langle \chi_U, \chi_V \rangle = \mathbf{1}_{U \simeq V} \quad \text{(indicator function)}.$$

1.3 Translating Between Characters and Representations

Throughout, $G : \mathbf{FinGroup}$ and $U, V : \mathbf{FinVect}^G$.

(T1) **Adjunction of induction to restriction**: For a subgroup $H \leq G$, the adjunction of functors $\mathrm{Ind}_H^G \dashv \mathrm{Res}_H^G$ appears as **Frobenius reciprocity**:

$$\mathbf{FinVect}^G(\mathrm{Ind}_H^G U, V) \simeq_{\mathrm{nat}.U,V} \mathbf{FinVect}^H(U, \mathrm{Res}_H^G V) \quad \rightsquigarrow \quad \langle \mathrm{Ind}_H^G U, V \rangle = \langle U, \mathrm{Res}_H^G V \rangle.$$

The inner product relation is a decategorification of the (categorical) adjunction.

(T2) **Equivalence of irreducible modules**: In the context where U, V are irreducible,

$$U \simeq V \iff \langle \chi_U, \chi_V \rangle = 1$$

and furthermore, in the same context,

$$U \not\simeq V \iff \langle \chi_U, \chi_V \rangle = 0.$$

See (G3) for a more concise statement.

(T3) **Frobenius reciprocity**: See (T1).

(T4) **Irreducible factors**: For U irreducible,

$$\text{a factor of } U \text{ appears in } V \iff \langle \chi_U, \chi_V \rangle \neq 0.$$

(A factor "appearing" means that $V \simeq U \oplus (\text{stuff})$.)

(T5) **Irreducibility**: See also (G3).

$$V \text{ is irreducible} \iff \langle \chi_V, \chi_V \rangle = 1.$$

2 The Story of Characters, Told in Broad Strokes

3 Proofs

We collect together proofs of all statements in the translation dictionary and in the "story told in broad strokes" (sections 1 and 2).

4 Some Character Tables